

**BAULKHAM HILLS HIGH SCHOOL
MATHEMATICS**

Year 12 Extension 2 Task 3

Thursday 27th May 2010

- Instructions:
- a) Write all your answers on your own paper.
 - b) Show all necessary working.
 - c) Marks may be deducted for careless or badly arranged work.

Time Allowed: 70 minutes

Question 1 (15 marks)

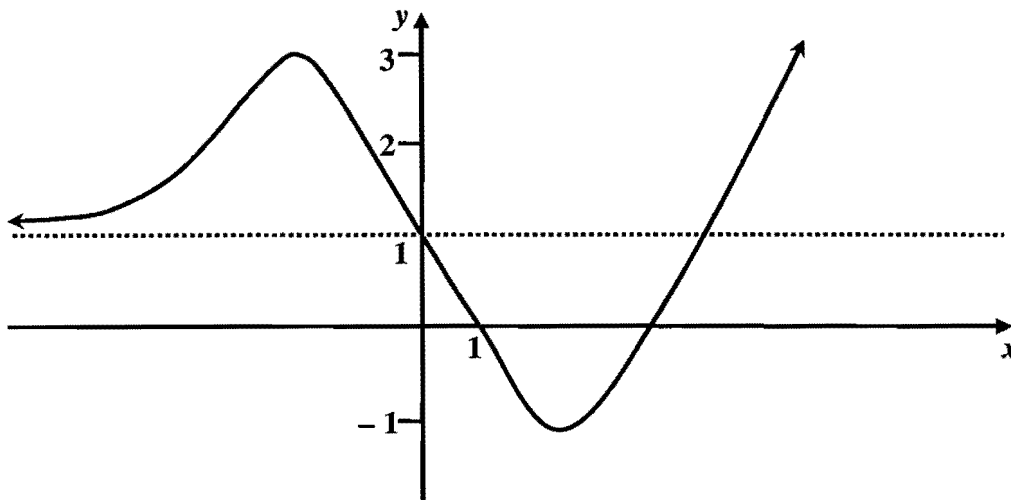
Marks

Find the following integrals:

- | | |
|--|---|
| (i) $\int \sin^4 x dx$ | 3 |
| (ii) $\int \frac{dx}{x^2 \sqrt{1+x^2}}$ | 3 |
| (iii) $\int x \tan^{-1} x dx$ | 3 |
| (iv) $\int \frac{x-1}{\sqrt{x^2+2x+3}} dx$ | 3 |
| (v) $\int \frac{5-2x}{(x-1)^2(x+2)} dx$ | 3 |

Question 2 (12 marks) Use a *separate* piece of paper

- a) The diagram shows the graph $y = f(x)$



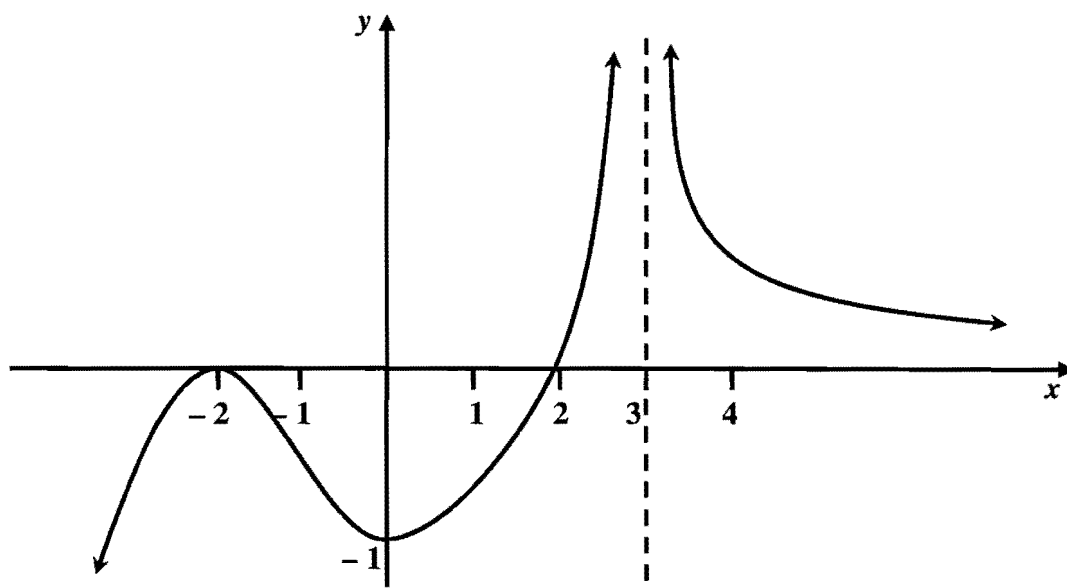
Draw separate one-third page sketches of the graphs of the following;

- | | |
|---------------------------|---|
| (i) $y = f(1-x)$ | 2 |
| (ii) $y = \frac{1}{f(x)}$ | 2 |
| (iii) $y = [f(x)]^2$ | 2 |
| (iv) $y = \tan^{-1} f(x)$ | 2 |

Question 2...continued**Marks**

- b) Given the graph of $y = f'(x)$ below, sketch the graph of $y = f(x)$

4

**Question 3 (13 marks)** Use a *separate* piece of paper

- a) $P\left(cp, \frac{c}{p}\right)$, $Q\left(cq, \frac{c}{q}\right)$ and $R\left(cr, \frac{c}{r}\right)$ are three points on the same branch of the

hyperbola $xy = c^2$

- (i) Show the equation of the tangent at P is $x + p^2y - 2cp = 0$ 2
- (ii) The tangents at P and Q intersect U . Find the coordinates of U 2
- (iii) The origin, U and R are collinear, find the relationship between p , q and r . 2
- b) (i) Prove that $\tan \frac{x}{2} \equiv \operatorname{cosec} x - \cot x$ 1
- (ii) Show that $\int \frac{dx}{1 + \cos x} = \operatorname{cosec} x - \cot x + c$ 3
- (iii) Hence, or otherwise, show that $\int \frac{x + \sin x}{1 + \cos x} dx = x \tan \frac{x}{2} + c$ 3

Question 4 (14 marks) Use a *separate* piece of paper

a) $P\left(3p, \frac{3}{p}\right)$ and $Q\left(3q, \frac{3}{q}\right)$ are points on the rectangular hyperbola $xy = 9$

- (i) Find the equation of the chord PQ . 2
- (ii) Find the coordinates of N , the midpoint of PQ . 2
- (iii) If the chord PQ is a tangent to the parabola $y^2 = 3x$, prove that the locus of N is $3x = -8y^2$ 3

b) Let $I_n = \int_1^e (\ln x)^n dx$, where n is a positive integer.

- (i) Show that $I_n = e - nI_{n-1}$ 2
- (ii) Let $J_n = \frac{I_n}{n!}$. Show that $\frac{1}{e}(1 + J_{10}) = \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots + \frac{1}{10!}$ 3
- (iii) It can be shown that $\sum_{r=2}^n \frac{(-1)^r}{r!} = \frac{1}{e} [1 + (-1)^n J_n]$ for all positive integers n . 2

Deduce the sum to infinity of the series $\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots$, justifying your answer carefully.

END OF EXAMINATION

Extension 2 Task 3 2010 Solutions

Question 1 (15)

(i) $\int \sin^4 x \, dx$

$$\begin{aligned} &= \frac{1}{8} \int (1 - \cos 2x)^2 \, dx \quad \checkmark \\ &= \frac{1}{8} \int (1 - 2\cos 2x + \cos^2 2x) \, dx \\ &= \frac{1}{8} \int (1 - 2\cos 2x + \frac{1}{2}(1 + \cos 4x)) \, dx \\ &= \frac{1}{8} \int (\frac{3}{2} - 2\cos 2x + \frac{1}{2}\cos 4x) \, dx \quad \checkmark \\ &= \frac{3}{16}x - \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + c \quad \checkmark \end{aligned}$$

(ii) $\int \frac{dx}{x^2\sqrt{1+x^2}}$ $x = \tan \theta$ $dx = \sec^2 \theta \, d\theta$

$$\begin{aligned} &= \int \frac{\sec^2 \theta \, d\theta}{\tan^2 \theta \sec \theta} \\ &= \int \frac{\sec \theta \, d\theta}{\tan^2 \theta} \\ &= \int \frac{1}{\cos \theta} \times \frac{\cos^2 \theta}{\sin^2 \theta} \, d\theta \\ &= \int \frac{\cos \theta}{\sin^2 \theta} \, d\theta \quad \checkmark \quad u = \sin \theta \quad du = \cos \theta \, d\theta \\ &= \int \frac{du}{u^2} \\ &= -\frac{1}{u} + c \\ &= -\operatorname{cosec} \theta + c \\ &= -\frac{\sqrt{1+x^2}}{x} + c \quad \checkmark \end{aligned}$$

(iii) $\int x \tan^{-1} x \, dx$ $u = \tan^{-1} x$ $v = \frac{1}{2}x^2$

$$\begin{aligned} &= \frac{1}{2}x^2 \tan^{-1} x - \frac{1}{2} \int \frac{x^2}{1+x^2} \, dx \quad \checkmark \\ &= \frac{1}{2}x^2 \tan^{-1} x - \frac{1}{2} \int [1 - \frac{1}{1+x^2}] \, dx \quad \checkmark \\ &= \frac{1}{2}x^2 \tan^{-1} x - \frac{1}{2}x + \frac{1}{2} \tan^{-1} x + c \quad \checkmark \end{aligned}$$

(iv) $\int \frac{x-1}{\sqrt{x^2+2x+3}} \, dx$

$$\begin{aligned} &= \frac{1}{2} \int \frac{(2x+2)}{\sqrt{x^2+2x+3}} \, dx - 2 \int \frac{dx}{\sqrt{(x+1)^2+2}} \quad \checkmark \\ &= \sqrt{x^2+2x+3} - 2 \log(x+1+\sqrt{x^2+2x+3}) + c \quad \checkmark \end{aligned}$$

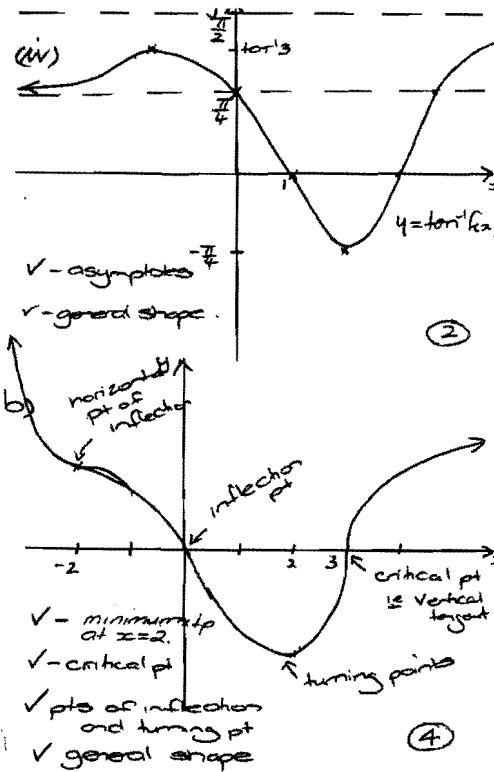
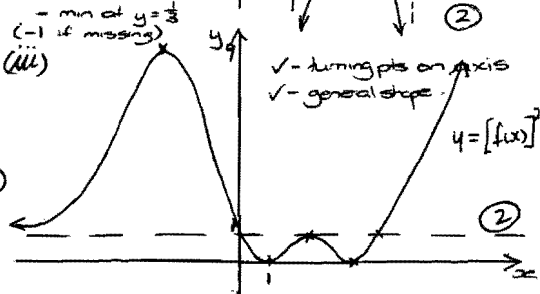
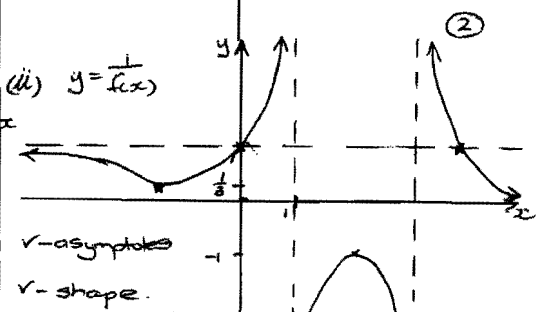
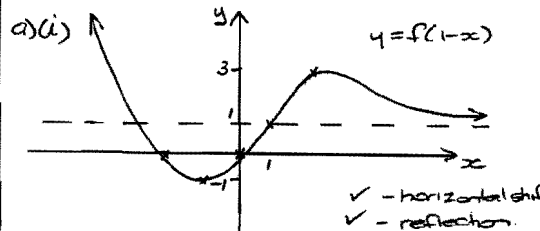
(v) $\int \frac{5-2x}{(x-1)^2(x+2)} \, dx$

$$\frac{A}{(x-1)} + \frac{B}{(x-1)^2} + \frac{C}{(x+2)} = \frac{5-2x}{(x-1)^2(x+2)}$$

$$\begin{aligned} A(x-1)(x+2) + B(x+2) + C(x-1)^2 &= 5-2x \\ \frac{x=1}{3B=3} \quad \frac{x=-2}{9C=9} \quad \frac{x=2}{4A+4B+C=1} \\ B=1 \quad C=1 \quad 4A+5=1 \\ 4A &= -4 \\ A &= -1 \quad \checkmark \end{aligned}$$

$$\begin{aligned} \int \frac{5-2x}{(x-1)^2(x+2)} \, dx &= \int \left[\frac{-1}{x-1} + \frac{1}{(x-1)^2} + \frac{1}{x+2} \right] dx \\ &= -\log(x-1) - \frac{1}{x-1} + \log(x+2) \quad \checkmark \\ &= \log\left(\frac{x+2}{x-1}\right) - \frac{1}{x-1} \quad \checkmark \end{aligned}$$

Question 2 (12)



$$\begin{aligned} x + \frac{2cp^2}{p+q} &= 2cp \\ x &= \frac{2cp^2 + 2cpq - 2cp^2}{p+q} \\ x &= \frac{2cpq}{p+q} \quad \checkmark \\ U & \text{ is } \left(\frac{2cpq}{p+q}, \frac{2c}{p+q} \right) \quad \textcircled{2} \end{aligned}$$

(iii) $m_{\text{av}} = m_{\text{or}}$ $\checkmark$ (or equiv)

$$\frac{\frac{2c}{p+q}}{\frac{2cpq}{p+q}} = \frac{c}{r}$$

$$\frac{1}{pq} = \frac{1}{r^2}$$

$$\underline{pq = r^2} \quad \checkmark \quad \textcircled{2}$$

b(i) $\operatorname{cosec} x - \cot x$

$$\begin{aligned} &= \frac{1+t^2}{2t} - \frac{1-t^2}{2t} \\ &= \frac{2t^2}{2t} \\ &= t \\ &= \tan \frac{x}{2} \quad \checkmark \quad \textcircled{1} \end{aligned}$$

Question 3 (13)

a) $xy = c^2$

$$y = \frac{c^2}{x}$$

$$\frac{dy}{dx} = -\frac{c^2}{x^2}$$

at $x = cp$, $\frac{dy}{dx} = -\frac{c^2}{c^2p^2} = -\frac{1}{p^2} \quad \checkmark$

$$y - \frac{c}{p} = -\frac{1}{p^2}(x - cp)$$

$$p^2y - cp = -x + cp \quad \checkmark$$

$$\underline{x + p^2y - 2cp = 0} \quad \textcircled{2}$$

(ii) tangent at Q: $x + q^2y - 2cq = 0$

$$\begin{aligned} x + p^2y &= 2cp \\ x + q^2y &= 2cq \\ (p^2 - q^2)y &= 2c(p - q) \\ y &= \frac{2c(p - q)}{(p - q)(p + q)} \\ y &= \frac{2c}{p + q} \quad \checkmark \end{aligned}$$

(ii) $\int \frac{dx}{1 + \cos x}$ $t = \tan \frac{x}{2}$ $dx = \frac{2dt}{1+t^2}$

$$\begin{aligned} &= \int \frac{\frac{2dt}{1+t^2}}{1 + \frac{1-t^2}{1+t^2}} \\ &= \int \frac{2dt}{2} \\ &= \int dt \\ &= t + c \\ &= \tan \frac{x}{2} + c \\ &= \operatorname{cosec} x - \cot x + c \quad \checkmark \quad \textcircled{3} \end{aligned}$$

OR

$$\begin{aligned} \int \frac{dx}{1 + \cos x} &= \int \frac{1 - \cos x}{1 - \cos^2 x} \, dx \\ &= \int \frac{1 - \cos x}{\sin^2 x} \, dx \\ &= \int [\operatorname{cosec}^2 x - \cot x \operatorname{cosec} x] \, dx \\ &= -\cot x + \operatorname{cosec} x + c \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad & \int \frac{x + \sin x}{1 + \cos x} dx \\ &= \int \frac{x dx}{1 + \cos x} + \int \frac{\sin x dx}{1 + \cos x} \quad \checkmark \\ & u = x \quad v = \operatorname{cosec} x - \cot x \\ & du = dx \quad dv = \frac{dx}{1 + \cos x} \\ &= x(\operatorname{cosec} x - \cot x) - \int (\operatorname{cosec} x - \cot x) dx \\ & \quad - \log(1 + \cos x) + C \\ &= x \tan \frac{x}{2} + \log(\operatorname{cosec} x + \cot x) \\ & \quad + \log(\sin x) + \log(1 + \cos x) \\ &= x \tan \frac{x}{2} + \log\left(\frac{1}{\sin x} + \frac{\cos x}{\sin x}\right) \\ & \quad + \log(\sin x) - \log(1 + \cos x) + C \\ &= x \tan \frac{x}{2} + \log(1 + \cos x) - \log(1 + \cos x) + C \\ &= \underline{x \tan \frac{x}{2}} \quad \checkmark \quad (3) \end{aligned}$$

$$\begin{aligned} \text{i.e. } 9p^2q^2 + 36(pt+q) &= 0 \\ p^2q^2 &= -4(pt+q) \quad \checkmark \\ 3x &= \frac{9(pt+q)}{2} \quad -8y^2 = -8\left(\frac{3(pt+q)}{2pq}\right)^2 \\ &= \frac{18(pt+q)^2}{p^2q^2} \\ &= \frac{-18(pt+q)^2}{-4(pt+q)} \quad \checkmark \\ &= \frac{9(pt+q)}{2} \\ \therefore \text{locus of } N \text{ is } 3x &= -8y^2 \quad (3) \end{aligned}$$

$$\begin{aligned} b) I_n &= \int_1^e (\log x)^n dx \\ u &= (\log x)^n & v &= x \\ du &= \frac{n(\log x)^{n-1}}{x} dx & dv &= dx \quad \checkmark \\ I_n &= \left[x(\log x)^n \right]_1^e - n \int_1^e (\log x)^{n-1} dx \quad \checkmark \\ &= \underline{\underline{e - n I_{n-1}}} \quad (2) \end{aligned}$$

Question 4 (14)

$$\begin{aligned} a) (i) m_{RR} &= \frac{\frac{3}{p} - \frac{3}{q}}{\frac{3p - 3q}{pq}} \\ &= \frac{\frac{q-p}{pq}}{\frac{p-q}{pq}} \\ &= -\frac{1}{pq} \end{aligned}$$

$$y - \frac{3}{p} = -\frac{1}{pq}(x - 3p)$$

$$\underline{x + pqy - 3(p+q) = 0} \quad \checkmark \quad (2)$$

$$(ii) \quad N\left(\frac{3p+3q}{2}, \frac{\frac{3}{p}+\frac{3}{q}}{2}\right) \checkmark$$

$$= \left(\frac{3p+3q}{2}, \frac{3p+3q}{2pq}\right) \checkmark \quad (2)$$

(iii) $y^2 = 3x$
 $x = \frac{1}{3}y^2$
 $\frac{1}{3}y^2 + pqy - 3(p+q) = 0$
 $y^2 + 3pqy - 9(p+q) = 0$
 If tangent then $\Delta = 0$ ✓

$$\begin{aligned} \text{(iii)} \quad & \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots \\ &= \sum_{r=2}^{\infty} \frac{(-1)^r}{r!} \\ &= \lim_{n \rightarrow \infty} \frac{1}{e} \left[1 + (-1)^n J_n \right] \\ &= \lim_{n \rightarrow \infty} \frac{1}{e} \left[1 + (-1)^n \frac{I_n}{n!} \right] \end{aligned}$$

as $n \rightarrow \infty$, $n! \rightarrow \infty$

for $n > 0, 0 < I_n < e$

$$\therefore \lim_{n \rightarrow \infty} \frac{H_n}{C} = 0$$

Thus

$$\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots = \underline{\underline{\frac{1}{e}}} \textcircled{2}$$

✓ answer

✓ justification

$$\begin{aligned}
 (ii) & \frac{1}{e} (1 + I_0) \\
 &= \frac{1}{e} (1 + \frac{I_0}{10!}) \\
 &= \frac{1}{e} (1 + \frac{e}{10!} - \frac{10I_9}{10!}) \quad \checkmark \\
 &= \frac{1}{e} (1 + \frac{e}{10!} - \frac{I_9}{9!}) \\
 &= \frac{1}{e} (1 + \frac{e}{10!} - \frac{e}{9!} + \frac{9I_8}{9!}) \quad \checkmark \\
 &= \frac{1}{e} (1 + \frac{e}{10!} - \frac{e}{9!} + \frac{I_8}{8!}) \\
 &= \frac{1}{e} (1 + \frac{e}{10!} - \frac{e}{9!} + \frac{e}{8!} - \dots + \frac{e}{2!} - \frac{e}{1!} + \frac{I_0}{0!}) \\
 &= \frac{1}{e} (1 + \frac{e}{10!} - \frac{e}{9!} + \frac{e}{8!} - \dots + \frac{e}{2!} - \frac{e}{1!} + \int dx) \\
 &= \frac{1}{e} (1 + \frac{e}{10!} - \frac{e}{9!} + \frac{e}{8!} - \dots + \frac{e}{2!} - e + [x]_1^e) \\
 &= \frac{1}{e} (1 + \frac{e}{10!} - \frac{e}{9!} + \frac{e}{8!} - \dots + \frac{e}{2!} - e + e - 1) \\
 &= \frac{1}{e} (\frac{e}{10!} - \frac{e}{9!} + \frac{e}{8!} - \frac{e}{7!} + \frac{e}{6!} - \frac{e}{5!} + \frac{e}{4!} - \frac{e}{3!} + \frac{e}{2!}) \\
 &= \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots + \frac{1}{10!} \quad \checkmark \quad (3)
 \end{aligned}$$